

Ishan Sinha

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EDUCATION

Michigan State University

Bachelor of Science in Computer Science | Concentration: Artificial Intelligence

East Lansing, MI

Aug. 2024 – Dec. 2027

- Minors: Data Science; Computational Math, Science & Engineering; Mathematics | **Michigander Scholar**

EXPERIENCE

Auto-Owners Insurance

Software Development Intern

Lansing, MI

May 2026 – Aug. 2026

- Optimized enterprise Guidewire Claim Center configurations to ensure high availability and reduce latency for large-scale claims.
- Enhanced backend user update workflows to aggregate relevant data, streamlining the visual presentation of user profiles for internal review.

MSU AI Club

Workshop Coordinator

East Lansing, MI

Aug. 2026 – Present

- Directed technical workshops on machine learning architectures, translating complex mathematical concepts like Bayesian probability into interactive Python models for audiences of 100-200 members.
- Managed core club operations and curriculum development, fostering an academic environment focused on emerging AI methodologies and compute infrastructure.

SEE-Insight Lab

Undergraduate Researcher

East Lansing, MI

Sep. 2026 – Present

- Architected knowledge representations and prompt logic for an AI Teaching Assistant, establishing quantitative evaluation rubrics to assess the accuracy and educational alignment of agentic responses.
- Conducted lab-wide code reviews and triaged GitHub pull requests, auditing legacy machine learning scripts to enforce software engineering best practices and maintain repository hygiene.

Facility for Rare Isotope Beams (FRIB)

Undergraduate Researcher

East Lansing, MI

Jan. 2026 – Sep. 2026

- Engineered a 3D sparse neural network framework, accelerating rare event detection in high-dimensional physical datasets.
- Optimized sparse tensor operations, significantly reducing compute latency for 3D structural analysis.

MSU Unmanned Systems

Software Project Manager

East Lansing, MI

Nov. 2025 – Present

- Directed autonomous flight software execution, deploying an NVIDIA Jetson Orin Nano edge-compute stack to actuate payloads via MAVLink telemetry.
- Engineered custom spherical math algorithms for spatial processing, resolving waypoint navigation to a 0.55-meter accuracy and securing 3rd place overall.

PROJECTS

OpticHash | Java (Spring Boot), C++, FastAPI, Docker, Nginx

- Architected a decoupled B2B microservices network using a Java Spring Boot API Gateway and Nginx proxy to stream multi-part payloads across multi-stage Docker containers.
- Authored a native C++ microservice utilizing OpenCV to evaluate structural geometry, generating O(1) Perceptual Hashes (pHash) that shield downstream ML nodes from redundant processing.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, JavaScript, SQL, HTML/CSS

Frameworks: PyTorch, Scikit-Learn, Hugging Face (RoBERTa, Whisper), Spring Boot, FastAPI, ChromaDB, OpenCV, TensorRT

Proficiencies: RAG pipelines, Microservices Architecture, Compute Optimization, Unsupervised Learning

Tools: Git, GitHub, Docker, Nginx, CI/CD, MAVLink, REST APIs, Linux/Unix